

Kovuru Harish Kumar Full Stack Developer

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Summary

Results-driven Full Stack Developer with 4+ years of experience in designing, developing, and deploying scalable web applications and microservices. Skilled in both frontend and backend technologies, cloud infrastructure, and real-time data streaming. Strong expertise in building responsive user interfaces, secure APIs, and distributed systems. Proficient in modern development tools and agile practices, with a focus on performance, reliability, and maintainability. Adept at collaborating across teams to deliver high-quality software solutions in fast-paced, enterprise environments.

Technical Skills

- Frontend:** React.js, Redux Toolkit, React Query, HTML5, CSS3, Bootstrap, Material-UI, D3.js, Recharts
- Backend:** Java, Spring Boot, Spring MVC, Spring Security (JWT, RBAC), Hibernate, RESTful APIs, Spring Batch
- Databases:** PostgreSQL (AWS RDS), MySQL, Snowflake
- Streaming & Messaging:** Apache Kafka, Websocket
- Search & Indexing:** OpenSearch
- Cloud & DevOps:** AWS (ECS, RDS, CloudWatch, ALB, Auto Scaling), Docker
- Testing & QA:** JUnit, Mockito, Cypress
- CI/CD & Deployment:** Dockerized microservices, load testing, penetration testing, integration testing
- Other Tools:** Git, Agile/Scrum methodologies

Professional Experience

Full Stack Developer, BMO Harris Bank

10/2024 – Present | Remote, USA

- Contributed to building a Customer Portfolio Intelligence Platform to help wealth managers monitor portfolios, assess asset allocations, and identify early risk indicators. Collaborated with product, compliance, data, and infrastructure teams to improve delivery efficiency by 30%.
- Designed and developed interactive dashboards using React, D3.js, and Recharts to visualize portfolio trends and risk scores. Implemented React Query for efficient data fetching and Redux Toolkit for state management, enhancing UI performance by 35%.
- Developed backend microservices using Spring Boot with JWT-based authentication and RBAC via Spring Security. Built services to manage portfolio data, connect with PostgreSQL on AWS RDS, and process historical insights through Snowflake and Spring Batch.
- Integrated Kafka for real-time market data and transaction streaming, enabling instant portfolio updates, automated alerts, and faster response to risk events across the system.
- Delivered robust testing and deployment pipelines by implementing unit, integration, and end-to-end tests with JUnit, Mockito, and Cypress. Authored technical documentation, and performed load, penetration, and integration testing—boosting system reliability and deployment success rate by 50%.
- Deployed Dockerized microservices to AWS ECS with Application Load Balancers and Auto Scaling. Improved reliability and observability with CloudWatch, increasing system uptime by 40% and minimizing manual deployment errors by 60%.

Software Engineer, Capgemini

01/2020 – 08/2023 | Telangana, India

- Designed and implemented a nomination and award management system for EdgeVerve, enabling employee self and peer nominations, automating approval workflows based on rating hierarchy, reducing manual processing time by over 60%.
- Developed responsive SPAs using React.js, Redux, HTML5, CSS3, Bootstrap, and Material-UI, creating reusable components and dynamic forms that improved page load speed, performance, and overall user engagement by approximately 40%.
- Built scalable backend microservices with Java, Spring Boot, Spring MVC, Hibernate, and RESTful APIs, enabling modular frontend integration and achieving 50% faster delivery of new features through independently deployable services.
- Integrated Kafka for asynchronous message streaming and OpenSearch for real-time data indexing and querying, ensuring rapid data retrieval and robust handling of event-driven operations within the distributed system.
- Designed and executed complex SQL queries, joins, and stored procedures in MySQL for handling mission-critical business logic and data operations, ensuring data integrity and performance optimization.
- Ensured high code quality and minimal defects by implementing unit and integration testing using JUnit and Mockito, consistently achieving over 90% test coverage and reducing production issues significantly.
- Managed AWS cloud infrastructure and introduced a retrievable mechanism for transient API failures like timeouts and service unavailability, enhancing fault tolerance, availability, and reliability across services.

Education

Master of Science in Computer Science

09/2023 – 06/2025

DePaul University, Chicago, IL, USA

Bachelor of Technology in Computer Science

06/2018 – 05/2021

VNR Vignana Jyothi Institute of Engineering and Technology, Telangana, India

Certifications

- Data Analytics, Cisco
- HackerRank Problem Solving (Intermediate) – Data Structures & Algorithms
- Applied Data Science with Python: from University of Michigan, Coursera